RoadWatch AP

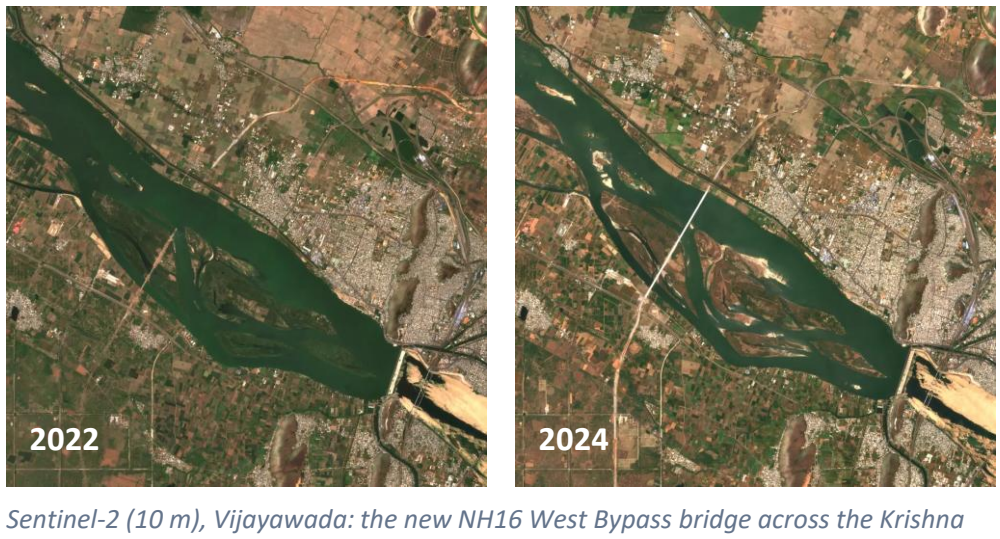
Automatic Road Extraction and New-Road Change Alerting from Multi-Temporal Satellite Images

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Department of CSE · KONERU LAKSHMAIAH EDUCATION FOUNDATION

Scan: live demo

SIH1564 (NTRO) · CLIMATE-06



Sentinel-2 (10 m), Vijayawada: the new NH16 West Bypass bridge across the Krishna appears between the two seasons.

The problem

Roads change every season, but maps are updated by hand — slowly and incompletely, so planning and monitoring work from out-of-date evidence.

Our goal

Read free satellite images, map every road and raise trustworthy alerts for genuinely new roads — each with before/after proof.

How it works

1 Composite
Median of the 8 least-cloudy passes per season

2 Segment
U-Net (ResNet) road probability, ONNX

3 Vectorise
Skeleton → road graph per season

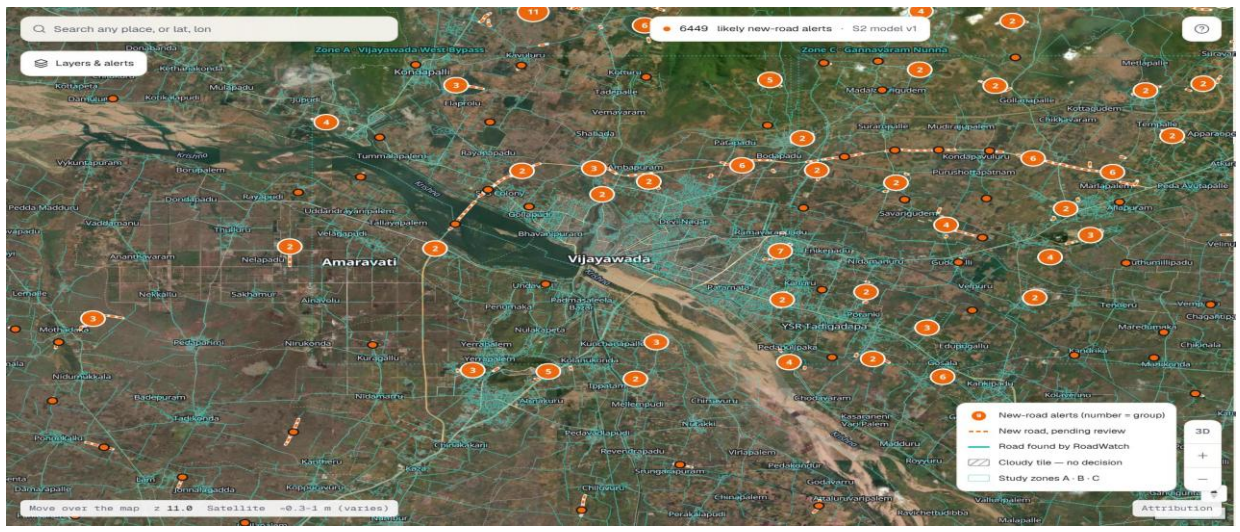
4 Compare
2022 baseline vs every later season

5 Confirm
Seen ≥ 3 later dates; cloudy = abstain

6 Alert
Score, stable id, flags, evidence

Alert score = model confidence × persistence × connectivity × data quality

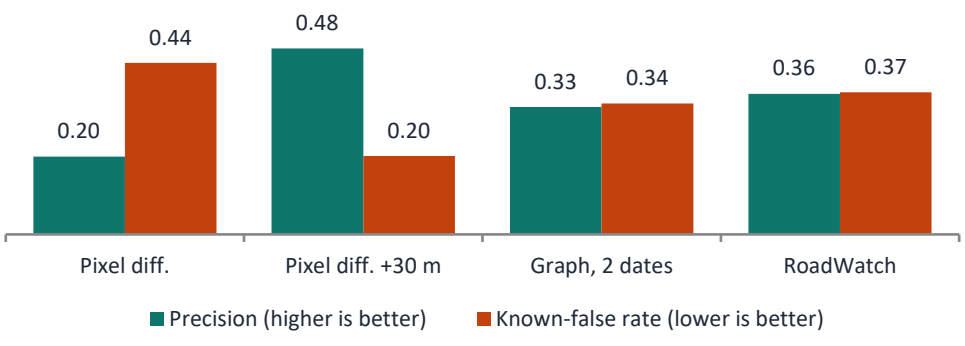
Results at scale — live at roadwatch-ap.onrender.com



Teal = roads extracted by RoadWatch · orange = new-road alerts

11,920	**alerts published** — AP · Tamil Nadu · Hyderabad · Bengaluru
6,449	**likely new roads** — 5,471 flagged as probably not new
2.9 lakh km	**roads extracted** — served as map vector tiles
98.7 %	**of South India mapped** — all-India map in progress

Change detection vs baselines (zones A–C)



Road extraction models





Conclusion

- Deployed end to end; 16× fewer alerts than pixel difference
- KPIs measured honestly; 14/14 negative tests pass
- Updates itself twice a year — no manual step

Future scope

- All-India map; multi-year training
- Sentinel-1 radar for monsoon months
- Field validation; reviews feed retraining